

GraphRAG for Enterprise Intelligence

Agentic Knowledge Graph Platform · Built for Client Delivery



Standard AI Fails Enterprise Knowledge

Hallucination at Scale

LLMs invent facts when knowledge is complex, contradictory or spread across hundreds of documents

No Reasoning Chain

Can't trace logic across connected entities — a contract links to a regulation links to a company

No Audit Trail

Clients in regulated industries need to know exactly where an answer came from and who authorised it

PwC clients in banking, regulatory, and audit need explainable, traceable, contradiction-free AI answers — not black boxes.

Three Layers. One Coherent System.

Knowledge Graph

Neo4j 5.x

- Entity resolution (4-stage)
- Ontology enforcement
- Forward-chaining inference
- Contradiction detection
- Bitemporal modeling

Retrieval Pipeline

6-stage hybrid

- Vector ANN (3072d Gemini)
- BM25 + RRF fusion
- Cross-encoder reranking
- Multi-hop graph traversal
- GAT/GCN GNN scoring

Agent Layer

Groq + Google ADK

- Tool invocation (FunctionTool)
- IRCoT iterative reasoning
- Session context (Redis)
- RAGAS evaluation
- JWT-secured REST API

Every JD Requirement — Demonstrated in Code

Role Requirement	What's Built & Where
✓ Neo4j + graph-based data modeling	39 KG modules · 572-line neo4j_client.py · production Cypher (UNWIND, COUNT{}, vector ANN, BM25)
✓ Graph RAG patterns	6-stage pipeline: vector → BM25+RRF → reranker → multi-hop traversal → GNN → LLM synthesis
✓ LLM integration + tool invocation	Google ADK with FunctionTool · Groq llama-3.3-70b · agentic IRCoT retriever
✓ Design + validate MVP solutions	304 passing tests · CI/CD (GitHub Actions) · Docker multi-stage · runbook · regulatory demo
✓ Agentic chatbot, autonomous tools	QueryAgent + IngestionAgent + EvaluationAgent · session context · Redis result store
✓ Secure integration	OAuth 2.0 · JWT scopes · rate limiting · per-tenant isolation · GDPR erasure

L I V E D E M O

Regulatory Compliance Intelligence

You will see:

1. Domain ontology loading from YAML — aerospace regulatory type hierarchy
2. Forward-chaining inference — AD-2024 transitively supersedes AD-2020
3. Contradiction detection — same aircraft IS_AIRWORTHY and IS_UNAIRWORTHY
4. Authority chain query — which document is the current governing authority?
5. Live REST API — ingest a document, query with session context

```
python scripts/demo_regulatory.py
```

Three PwC Client Scenarios

01

Regulatory Intelligence

Banking / Insurance clients

Ingest 10,000 regulatory documents. Detect when a new directive supersedes an old one. Surface contradictions before they reach an audit. Show exactly which document mandates a given control.

Document authority · Inference · Contradiction detection

02

Audit Knowledge Base

Audit & Assurance teams

Link entities across evidence documents. Multi-hop traversal finds connections human reviewers miss. Agentic re-search iterates until an answer is grounded in source material.

6-stage RAG · IRCoT fallback · Session context

03

Compliance Monitoring

Risk & Regulatory consulting

Temporal knowledge modeling tracks when facts changed. Authority-weighted confidence scores flag low-quality sources. GDPR-compliant entity erasure on request.

Bitemporal · Confidence calibration · GDPR

Production-Grade. Not a Prototype.

22,650

Lines of production Python

304

Passing tests (unit + integration)

39

Knowledge graph modules

3

ADRs: graph, inference, confidence

Knowledge Graph Capabilities

- Forward-chaining inference (Datalog rules)
- OWL-RL reasoning + SPARQL bridge (rdflib)
- TransE link prediction + entity embeddings
- Leiden community detection (graspologic)
- Bitemporal: valid time + transaction time

Production & Operations

- CI/CD — GitHub Actions · Docker multi-stage
- OAuth 2.0 + JWT + rate limiting + tenant isolation
- GDPR erasure cascade + PII detection
- RAGAS evaluation (faithfulness, recall, precision)
- Runbook, ADRs, contributing guide

READY TO DELIVER

Built for Client Work.

Day-one delivery

Complete platform ready — no ramp-up on fundamentals

Regulatory domain expertise

Aerospace compliance demo runs today; maps to banking, audit, and insurance use cases

Open-source proof

Every claim is verifiable in code — grep for the function, run the test, see it working